

Obtain protein annotations

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Here, we obtain uniprot details and GO annotations for the proteins quantified. To save this needing to be re-run if there are changes in how the output is processed to protein-level abundances, we will obtain the feature data for all proteins detected in the evidence files.

```
maxQuant_outfiles <- Sys.glob(here('raw/*/evidence.txt*'))
```

```
all_proteins <- maxQuant_outfiles %>% lapply(function(x){
  read.delim(x) %>%
  pull(Leading.razor.protein) %>%
  unique() %>%
  strsplit(';') %>%
  unlist() %>% unique()
}) %>% unlist() %>% unique()

length(all_proteins)
```

```
## [1] 9612
```

Uniprot protein details

Below, we obtain the protein details from Uniprot using the uniprotREST package.

```
uniprot2details <- uniprot_map(
  all_proteins,
  from='UniProtKB_AC-ID',
  method='stream') %>%
dplyr::rename('UniprotID'=From) %>%
select(-Entry) %>%
mutate(Gene.Names.First=gsub(' .*', '', Gene.Names))
```

```
## Running job: f0lzF2yOXf
```

```
## Checking job status...
```

```
## Job complete!
```

```
## Downloading: 0 B      Downloading: 0 B      Downloading: 0 B      Downloading: 0 B      Downloading: 0 B
```

```
# Some UniprotIDs generate multiple rows. It looks like this is because they have been subsequently 'detected'
```

```
uniprot2details_unique <- uniprot2details %>%
  group_by(UniprotID) %>%
  summarise_all(.funs=function(x) paste(x, collapse=';'))

saveRDS(uniprot2details_unique, here('external/uniprot2details_unique.rds'))
```

GO terms

Below we obtain the GO terms from Uniprot using the API and then we reformat the output before expanding the GO terms to include all parent terms too.

```
tempfilename <- tempfile()
```

```
uniprot_map(  
  ids = all_proteins,  
  from = "UniProtKB_AC-ID",  
  method='stream',  
  to = "UniProtKB", compressed = TRUE,  
  format = "tsv", path = tempfilename,  
  fields = "go",  
)
```

```
## Running job: f0lzF2yOXf
```

```
## Checking job status...
```

```
## Job complete!
```

```
## Downloading: 0 B      Downloading: 0 B      Downloading: 0 B      Downloading: 0 B      Downloading: 0 B
```

```
go_res <- read.delim(tempfilename) %>% dplyr::rename(c('UNIPROTKB'='From'))
```

```
file.remove(tempfilename)
```

```
## [1] TRUE
```

```
# The Uniprot API returns a data.frame with all the GO terms for each protein in a single column  
# Below, we separate each GO terms to be on a different row
```

```
go_res <- go_res %>%  
  filter(Gene.Ontology..GO.!='') %>% separate_rows(Gene.Ontology..GO., sep = '; ') %>%  
  separate_wider_delim(Gene.Ontology..GO., ' [GO:', names=c('GO_desc', 'GO.ID')) %>%  
  mutate(GO.ID=paste0('GO:', gsub(']'$', '', GO.ID)))
```

```
# And then expand the GO terms so that all ancestor/parent terms are annotated to each protein  
# This is to avoid issues with GO over-representation testing packages only using  
# the directly annotated terms and not the parent terms
```

```
go_res_all <- get_ancestor_go(go_res)
```

```
## [1] "Getting all ancestor GO terms for 14722 observed terms. This may take a while!"
```

Save to file for use in other notebooks.

```
saveRDS(go_res_all, here('external/go_res_all.rds'))
```